



Predicting Incidence of Some Critical Events by Sun Signs—The PISCES Study

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Background: Zodiac signs are believed to influence personality and health. Those born under the sun sign Leo are alleged to be big-hearted and at risk for heart disease.

Methods: We compared the demographic and exercise variables among a cohort of 32,386 patients undergoing stress testing at a large academic center. Further, we studied the association of Leo sun sign with long-term mortality.

Results: There were only minor differences in the baseline or exercise variables among Leos and the remainder of the cohort. There was a total of 2586 deaths over a period of 5 years. There was a slight excess of deaths among the Leos (9.6% vs. 8.7%). This relationship acquired a borderline significance after multivariate adjustment (hazard ratio, 1.17; 95% confidence interval [CI], 1.03 to 1.33, $p=0.019$). However, no significant difference in risk of mortality was seen in a propensity-adjusted model (adjusted hazard ratio, 1.10; 95% CI, 0.92 to 1.31; $p=0.30$).

Conclusion: Being a Leo is not associated with any adverse cardiac risk. Our findings should provide reassurance to the large population of Leos that together make up approximately one twelfth of mankind.

Astrology is a widely popular pastime with a large following worldwide. The zodiac sign that one is born under allegedly influences multiple aspects of human behavior and personality, interpersonal interactions and health. Zodiac signs are determined by date of birth and once acquired are not amenable to change. Although their use is immensely popular, there are limited data on the health implication of zodiac signs. Because time of birth is somewhat predictable, such an association would have significant preventive health implications. According to the American Federation of Astrologers, the sun sign Leo is associated with a big heart and predisposes its bearer to back and heart trouble.

Therefore, our *a priori* hypothesis was that subjects born under the sun sign Leo would be more likely to have abnormal stress tests and would be more likely to be at risk of dying during follow-up. Further, because Leos are allegedly big-hearted, we studied the association of being born under the sun sign Leo and left ventricular hypertrophy (LVH) as defined by electrocardiogram (ECG) criteria.

Methods

Our cohort included 32,386 patients who underwent exercise stress testing at Cleveland Clinic stress laboratories between the years 1990 and 1999. We excluded patients with atrial fibrillation and known valvular disease. The stress-testing protocol in our laboratory has been described elsewhere in detail. Data on clinical presentation, known coronary risk factors, medication use and prior cardiac and noncardiac diagnosis were obtained from a structured patient interview and chart review and prospectively entered into an online computerized database. Electrocardiographic hypertrophy was assessed by Solcolow-Lyon criteria. Symptom-limited maximal exercise testing was performed using standardized exercise test protocols under the supervision of an exercise physiologist and/or a cardiology fellow. Heart rate, blood pressure, rhythm, symptoms, estimated workload and ECG data were recorded online. Patients were encouraged to continue to exercise even after age-appropriate target heart rate was achieved. A failure to use 80% of the heart-rate reserve (the difference between the maximal achievable heart rate [220 beats per minute minus age in years] and the resting heart rate) was considered evidence of an impaired chronotropic response. Heart rate recovery, defined as the difference in heart rate at 1 minute after cessation of exercise from the heart rate at peak exercise, was considered abnormal if it was ≤ 12 bpm for standard exercise testing and ≤ 18 bpm for patients who underwent stress echocardiography.

Zodiac signs were defined per the American Federation of Astrologers; the sign of interest, Leo, was defined as patient birth date between July 23 and August 22 (<http://www.astrologers.com/>).

Vital status was assessed for all patients using the Social Security Death Index. This index previously has been shown to be highly specific and unbiased. Further, the sensitivity of this index in our stress test population has been previously validated. Our primary end point for this study was all-cause mortality. All-cause mortality is an objective and unbiased end point, unlike cardiac or noncardiac mortality, which may be significantly impacted by ascertainment bias.

Continuous variables were expressed as mean $\pm$ SD, and discrete variables were expressed as frequency counts and percentages. The association of zodiac signs with mortality was assessed using proportional hazard regression. Unadjusted and age- and gender-adjusted hazard of mortality associated with being a Leo was calculated. Leo sun sign was forced into a multivariable model of known predictors of mortality in this population: age; gender; poor physical fitness; resting heart rate; abnormal heart rate recovery; chronotropic incompetence; ST abnormality during stress test; nondiagnostic ST changes; presence of abnormal Q waves or left or right bundle branch block on resting ECG; ECG evidence of LVH; history of smoking within the past year; known history of coronary artery disease; history of

Table 1. Demographic and Clinical Characteristics of the Cohort According to Presence or Absence of Leo Zodiac

Baseline characteristics	Non-Leo (n = 29,534)	Leo (2852)
Age (mean, in years)	56.29	56.26
Male sex	20,841 (70.6%)	819 (71.3%)
Current or recent smoker	5274 (17.9%)	507 (17.8%)
Diabetes mellitus (on insulin)	1099 (3.7%)	108 (3.7%)
Diabetes mellitus (no insulin)	2398 (8.1%)	213 (8.8%)
Prior coronary artery disease	10,092 (34.2%)	952 (33.4%)
History of congestive heart failure	1173 (4.0%)	110 (3.9%)
Prior coronary artery bypass grafting	4211 (14.3%)	426 (14.9%)
Prior percutaneous coronary intervention	3242 (11.0%)	303 (10.62%)
Lipid lowering therapy	3927 (13.3%)	373 (13.1%)
ACE inhibitor use	3579 (12.1%)	343 (12.0%)
Beta-blocker use	5243 (17.8%)	530 (18.5%)
Aspirin use	9617 (32.6%)	943 (33.1%)
Digoxin use	1374 (4.7%)	131 (4.6%)
Right bundle branch block	2186 (7.4%)	175 (6.1%)
Left bundle branch block	482 (1.6%)	46 (1.6%)
LVH	548 (1.9%)	56 (2.0%)
Q waves on ECG	2973 (10.0%)	262 (9.2%)
Resting tachycardia	1351 (4.6%)	116 (4.1%)
Chronic obstructive pulmonary disease	861 (2.9%)	84 (2.9%)
End-stage renal disease	239 (0.8%)	27 (0.9%)

heart failure; use of aspirin, digoxin, angiotensin-converting enzyme inhibitors or beta-blockers; history of chronic obstructive lung disease; history of diabetes mellitus; insulin use and history of end-stage renal disease.

Because standard multivariable regression may fail to adequately adjust for multiple confounders, we used propensity analysis in a three-step process. First, a nonparsimonious logistic regression model was constructed to generate a propensity score for those of Leo sun sign, with all the variables presented in Tables 1 and 2 considered as covariates. Second, a matching macro was used to match Leo patients to unique non-Leo patients according to propensity score. Third, proportional hazards modeling was performed in the propensity-matched population, with and without adjustment for the propensity score and covariates.

A p value of <0.05 was used as the criterion for statistical significance. All analyses were performed using the SAS 8.1 statistical package (SAS, Inc., Cary, NC).

Table 2. Exercise Characteristics According to Presence or Absence of Leo Zodiac

Exercise characteristics	Non-Leo	Leo
Poor physical fitness	8835 (29.9%)	848 (29.8%)
Chronotropic incompetence (no β -blocker)	5670 (19.2%)	531 (18.6%)
Abnormal heart rate recovery	7136 (24.2%)	701 (24.6%)
Angina, non test limiting	3119 (10.6%)	311 (10.9%)
Angina, test limiting	241 (0.8%)	22 (0.8%)
Abnormal ST-segment changes (1–2 mm)	3330 (11.2%)	303 (11.0%)
Abnormal ST-segment changes (>2 mm)	679 (2.3%)	79 (2.7%)
Abnormal ST-segment changes (non-diagnostic)	5793 (19.6%)	548 (19.2%)

Results

Of 32,386 patients, 2852 (9%) had a Leo sun sign. Table 1 describes the demographic and clinical characteristics of this cohort according to presence or absence of Leo sun sign. Of note, Leo sun sign was not associated with any marked increased likelihood of ECG LVH or with other ECG abnormalities. There were only minor associations between Leo zodiac and prior cardiac history.

Table 2 presents the key stress test findings in Leos as compared with the rest of the cohort. Again, only minor differences were seen in any of the key variables that are routinely studied in our stress laboratory and are known predictors of outcome in the stress test population.

During a mean follow-up of 5 years, there were 2586 deaths. Patients with a Leo sun sign were at slightly increased risk of death (9.6% vs. 8.7%; age- and gender-adjusted hazard ratio, 1.13; 95% confidence interval, 0.99 to 1.29; $p=0.07$). After further adjusting for all the covariates listed in Tables 1 and 2 in a stepwise Cox proportional hazards model, a Leo sun sign emerged as an independent predictor of death (adjusted hazard ratio, 1.17; 95% confidence interval, 1.03 to 1.33; $p=0.019$).

In a cohort that consisted of the 2852 Leo patients and 2852 propensity-matched non-Leo patients, there were 509 deaths. There was no association between Leo zodiac and death (8.7% vs. 9.1%; age- and gender-adjusted hazard ratio, 0.91; 95% confidence interval, 0.77 to 1.09; $p=0.31$). After adjusting further for the propensity score and covariates, Leo sun sign remained unassociated with death risk (adjusted hazard ratio, 1.10; 95% confidence interval, 0.92 to 1.31; $p=0.30$).

Discussion

Zodiac, meaning “circle of animals,” refers to an imaginary belt extending 9° on either side of the plane of the Earth’s orbit. The 12 astrological signs of the zodiac are each considered to occupy 30° of this circle. These are grouped further into fire signs (Aries, Sagittarius, Leo), water signs (Cancer, Scorpio, Pisces), air signs (Libra, Aquarius, Gemini), and earth signs (Capricorn, Taurus, Virgo). Each sign is allegedly associated with key personality traits and predisposition to illness afflicting certain key body parts.

Previous data on the association of health and zodiac signs are limited to a subgroup analysis of the International Study of Infarct Survival (ISIS)-2 trial in which patients born under Gemini or Libra demonstrated a slightly adverse effect of aspirin on mortality, whereas patients born under the remainder of the zodiac signs had a significant mortality reduction with aspirin. This particular analysis was aimed at exposing the limitations inherent in retrospective subgroup analysis and not on exploring the relation between zodiac and health outcome. We believe our study is the first to explore the relationship between zodiac signs and health outcome in a large, well-defined cohort.

We did not find any correlation between sun signs and abnormality on stress testing. Further, we focused on subjects born under the sun sign of Leo, because they are alleged to have big hearts and are prone to weakness of the heart and the back. A big heart, although carrying favorable connotations in lay terms, is associated with a worse long-term outcome. Evidence of hypertrophy on ECG, although not highly sensitive, is an easily available marker of left ventricular hypertrophy and is associated with an adverse long-term outcome. Fortunately for Leos, only weak associations were identified.

Despite the absence of any marked differences in baseline, ECG or exercise characteristics, Leo sun sign did emerge as an independent predictor of mortality after standard multivariable adjustment. After careful consideration of confounding, however, by means of propensity scoring and matching, any such association completely disappeared. This analysis might serve as an example of the dangers of data torture when employing standard multivariable regression techniques in very large datasets and the potential value of propensity analysis for unmasking such torture.

Thus, our study failed to find an association between signs of zodiac and either stress test abnormalities or mortality. Further, those born under the sign of Leo do not appear to be at an increased hazard of long-term mortality above and beyond that of the general population. Our data should provide reassurance to Leos, who make up roughly 8 to 9% of the general population. Further, prospective parents attempting the notoriously difficult task of timing the birth of their progeny need not consider the zodiac for its health implications.

Suggested Reading

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